

Final Project Report

Crowd Detection and Crowd Level Classification Using YOLOv8



COMP6826001 – Deep Learning

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Abstract

Monitoring crowd density in indoor CCTV environments is an important task for safety management and operational planning, yet it remains challenging due to high crowd density, occlusion, and limited camera perspectives. This project proposes a deep learning based crowd detection system using the YOLOv8 object detection framework to estimate the number of individuals and classify crowd density levels in indoor public spaces. The Mall Dataset is employed as it provides annotated indoor CCTV scenes with varying crowd densities; head-point annotations are converted into bounding boxes to enable head-based person detection, which is more robust than full-body detection in dense crowd scenarios. The YOLOv8 model is fine-tuned from pretrained weights to adapt to the specific visual characteristics of the dataset. Experimental results show that the proposed approach achieves strong performance, with an mAP@0.5 of approximately 0.94, precision of 0.96, and recall of 0.90 on the test set, demonstrating the effectiveness of fine-tuning for accurate crowd detection and density classification in indoor CCTV applications.

1. Introduction

1.1 Background

Monitoring crowd conditions in indoor public spaces is an important aspect of safety management, facility planning, and operational efficiency. Environments such as shopping malls, stations, and public halls are typically equipped with closed-circuit television (CCTV) systems to observe human activities and crowd flow. However, manually monitoring crowd density through CCTV footage is inefficient and prone to human error, especially as crowd density increases, motivating the need for automated computer vision-based solutions (Bhuiyan et al., 2022).

From a computer vision perspective, crowd detection in indoor CCTV environments presents significant challenges. High crowd density often leads to severe occlusion, where individuals partially or fully block one another. In addition, fixed camera viewpoints and limited spatial resolution cause people to appear small and closely packed in the image. These factors significantly reduce the reliability of traditional full-body object detection approaches, particularly in dense indoor scenes (Bhuiyan et al., 2022).

Recent advances in deep learning, particularly convolutional neural network-based object detection models, have shown promising results in addressing complex visual recognition tasks, including crowd analysis. Among these models, the YOLO (You Only Look Once) family has gained attention due to its ability to perform fast and accurate object detection in a single forward pass, enabling real-time and near real-time applications (Sukkar et al., 2021; Gündüz & Işık,

2023). Consequently, applying a deep learning–based YOLO approach to crowd detection in indoor CCTV environments is both practically relevant and technically meaningful.

1.2 Problem Statement

Despite the availability of CCTV systems, accurately detecting individuals and estimating crowd density in indoor environments remains a challenging problem. High crowd density, overlapping individuals, and partial occlusion often degrade detection performance. In addition, many public datasets for crowd analysis do not provide bounding box annotations suitable for modern object detection models, further complicating the training process.

This project addresses the problem of how to effectively detect individuals and estimate crowd density in indoor CCTV images under crowded conditions, using a deep learning based object detection approach that can generalize to unseen frames without relying on manual annotations.

1.3 Objectives

The objectives of this project are as follows:

1. To develop a deep learning based crowd detection system for indoor CCTV environments using the YOLOv8 object detection framework.
2. To convert head-point annotations from the Mall Dataset into bounding box annotations compatible with YOLO training.
3. To fine-tune a pretrained YOLOv8 model to adapt to dense indoor crowd scenes.
4. To evaluate the performance of the proposed model using standard object detection metrics, including mAP@0.5, precision, and recall.
5. To implement an application that performs crowd counting and classifies crowd density levels into low, medium, and high categories.

1.4 Significance

This project is significant from both academic and practical perspectives. Academically, it demonstrates the application of modern deep learning techniques to a challenging real-world problem involving dense crowds and occlusion. The project also highlights the importance of data preprocessing, particularly the transformation of head-point annotations into bounding boxes for object detection tasks.

From a practical standpoint, the proposed system provides a foundation for automated crowd monitoring in indoor public spaces. By estimating crowd density and categorizing crowd levels in an interpretable manner, the system can support safety monitoring, facility management, and operational decision-making. Furthermore, the integration of the trained model into a deployable

application illustrates how deep learning models can be transitioned from experimentation to real-world usage.

2. Related Work

Recent studies show that deep learning has become the dominant approach for crowd analysis and surveillance applications. A comprehensive review by Bhuiyan et al. (2022) highlights that traditional computer vision methods struggle in dense crowd scenarios due to occlusion, scale variation, and low image quality, especially in CCTV-based environments. Their survey emphasizes that modern crowd analysis systems increasingly rely on deep neural networks to handle complex visual patterns in crowded scenes, including people detection, counting, and abnormal behavior analysis. This shift demonstrates the importance of deep learning based object detection frameworks for real-world crowd monitoring applications.

Several works focus specifically on applying YOLO-based architectures for pedestrian and crowd detection due to their real-time performance. Sukkar et al. (2021) demonstrated that YOLO models are effective for real-time pedestrian detection, addressing challenges such as illumination changes, occlusion, and background complexity. Similarly, Gündüz and Işık (2023) proposed a YOLO-based method for real-time crowd detection in indoor environments, showing that YOLO models achieve competitive accuracy and high frame rates when detecting and counting people in surveillance videos. These studies confirm that YOLO is well suited for real-time crowd detection tasks where both speed and accuracy are critical.

However, detecting full-body pedestrians in dense crowds remains challenging due to severe occlusions. To address this issue, recent research has explored head-based representations as a more robust alternative. Wu et al. (2025) proposed a head-anchor enhanced detection framework, showing that head regions are less prone to occlusion than full-body regions in crowded scenes. Their findings indicate that head-based detection improves robustness in high-density environments, even though head regions contain fewer discriminative features. This supports the use of head-based detection strategies for crowded surveillance scenarios, particularly when full-body visibility is limited.

In addition to detection, advances in YOLO architectures have demonstrated the flexibility of the framework for related human-centered tasks. The YOLO-Pose approach proposed by Maji et al.(2022). shows that YOLO-based models can jointly perform detection and localization tasks in crowded scenes within a single forward pass. By leveraging YOLO's end-to-end design, their work highlights how object detection frameworks can be adapted to handle complex human-related tasks efficiently. Motivated by these findings, this project adopts a YOLO-based approach and fine-tunes the model for head-based crowd detection using CCTV images, aligning with existing research that emphasizes robustness, real-time performance, and deployability in dense indoor environments.

3. Methodology

3.1 Dataset

3.1.1 Mall Dataset Overview

This research utilizes the Mall Dataset, a widely recognized benchmark for crowd counting and pedestrian detection in surveillance scenarios (Chen et al., 2012). The dataset was collected from a publicly accessible shopping mall using a fixed surveillance camera positioned at an elevated vantage point, providing an overhead view of pedestrian traffic patterns.

The dataset contains 2,000 image frames with a resolution of 640×480 pixels, representing various crowd densities. Each image corresponds to a single frame extracted from a continuous video stream, enabling frame-by-frame crowd analysis. The dataset provides head-point annotations, where each individual is labeled by a single point corresponding to the center of the head. This annotation strategy is well suited for dense crowd environments, as the head is the most consistently visible body part under occlusion.

3.1.2 Dataset Split

To ensure robust model evaluation and prevent overfitting, the dataset was partitioned using stratified random sampling with a fixed random seed (seed = 42) for reproducibility:

- **Training Set:** 1,400 frames (70%) - Used for model parameter optimization
- **Validation Set:** 400 frames (20%) - Used for hyperparameter tuning and early stopping
- **Test Set:** 200 frames (10%) - Used for final model evaluation

The random shuffling prior to splitting prevents temporal correlation bias, as consecutive video frames tend to be highly similar. This ensures that the model learns generalizable patterns rather than memorizing sequential dependencies.

3.2 Data Preprocessing

3.2.1 Annotation Conversion

The Mall Dataset provides annotations as (x,y) coordinates representing the centroid of each person's head. These point annotations needed to be converted to bounding boxes compatible with the YOLO object detection format.

3.2.2 Bounding Box Generation

For each person annotation point (x, y) , a rectangular bounding box was created with fixed dimensions:

- **Width:** 32 pixels
- **Height:** 48 pixels

These dimensions were chosen based on the average person size in the dataset's perspective view. The bounding box coordinates were calculated as:

$$\begin{aligned}x_1 &= x - \text{BOX_W}/2 & x_2 &= x + \text{BOX_W}/2 \\ y_1 &= y - \text{BOX_H}/2 & y_2 &= y + \text{BOX_H}/2\end{aligned}$$

Boundary clipping was applied to ensure bounding boxes remained within image dimensions:

$$\begin{aligned}x_1 &= \max(0, \min(x_1, W-1)) & x_2 &= \max(0, \min(x_2, W-1)) \\ y_1 &= \max(0, \min(y_1, H-1)) & y_2 &= \max(0, \min(y_2, H-1))\end{aligned}$$

where $W = 640$ and $H = 480$ represent the image width and height respectively.

3.2.3 YOLO Format Conversion

YOLO requires annotations in a specific format with normalized coordinates. Each bounding box was converted using:

$$\begin{aligned}c_x &= (x_1 + x_2) / (2 \times W) \\ c_y &= (y_1 + y_2) / (2 \times H) \\ w &= (x_2 - x_1) / W \\ h &= (y_2 - y_1) / H\end{aligned}$$

Where:

- **c_x, c_y :** Normalized center coordinates of the bounding box $\in [0,1]$
- **w, h :** Normalized width and height $\in [0,1]$
- **W, H :** Image width (640) and height (480)

The final YOLO annotation format for each frame is a text file containing one line per person:

```
class_id c_x c_y w h
```

where `class_id = 0` represents the 'person' class.

3.2.4 Data Augmentation

YOLOv8 applies built-in data augmentation during training to improve model generalization:

- **Mosaic Augmentation:** Combines 4 training images into one, forcing the model to learn objects in different contexts and scales
- **Random Resizing:** Images scaled to 640×640 pixels with aspect ratio preservation
- **Color Jittering:** Random adjustments to hue, saturation, and value (HSV) to handle lighting variations
- **Random Flip:** Horizontal flipping with 50% probability

3.3 Model Architecture

3.3.1 YOLOv8 Overview

This project employs YOLOv8n (nano variant), the latest iteration in the YOLO series developed by Ultralytics. YOLOv8 represents significant architectural improvements over previous versions, offering better accuracy-speed tradeoffs for real-time object detection applications.

The YOLOv8n variant was chosen for its optimal balance between computational efficiency and detection accuracy, making it suitable for real-time crowd monitoring applications with limited computational resources.

3.3.2 Architecture Components

1. Backbone Network

YOLOv8 utilizes a modified CSPDarknet53 backbone with the following features:

- **C2f modules:** Cross-Stage Partial Bottleneck with 2 convolutions, enabling efficient gradient flow and feature reuse
- **SPPF (Spatial Pyramid Pooling Fast):** Aggregates features at multiple scales using sequential max pooling operations
- **SiLU activation:** Smooth activation function $f(x) = x \cdot \text{sigmoid}(x)$ providing better gradient properties than ReLU

2. Neck Network

The Path Aggregation Network (PANet) structure facilitates multi-scale feature fusion:

- ❖ **Top-down pathway:** Propagates strong semantic features from deeper layers to enhance high-level understanding
- ❖ **Bottom-up pathway:** Enhances localization information from shallow layers for precise bounding box prediction
- ❖ **Feature Pyramid:** Generates multi-scale feature maps (P3, P4, P5) for detecting objects at different scales

3. Detection Head

YOLOv8 introduces an anchor-free, decoupled head architecture that improves detection flexibility:

- **Anchor-free design:** Predicts object centers directly without predefined anchor boxes, reducing hyperparameter complexity
- **Decoupled head:** Separate convolutional branches for classification and box regression tasks, allowing task-specific optimization
- **Distribution Focal Loss (DFL):** Models bounding box locations as probability distributions for more accurate predictions

The head produces predictions at three scales (8×, 16×, 32× downsampling) to handle objects of varying sizes, from small distant people to larger foreground individuals.

3.4 Training Setup

3.4.1 Transfer Learning Strategy

The model employs transfer learning by initializing with weights pre-trained on the COCO dataset, which contains 118,000 images across 80 object categories including persons. This approach provides several advantages:

- **Faster convergence:** Pre-trained features significantly reduce training time from days to minutes
- **Better performance:** Leverages robust visual representations learned from diverse data
- **Reduced overfitting:** Pre-trained weights provide strong initialization even with limited domain-specific training data

3.4.2 Hyperparameters

The following hyperparameters were configured for optimal training performance:

Hyperparameter	Value
Epochs	50
Batch Size	16
Image Size	640 x 640 pixels
Learning rate	0.01
Optimizer	AdamW (default YOLOv8)
Early Stopping	Patience = 10 epochs
Hardware	NVIDIA A100 GPU with 40GB VRAM

Table 1: Training Hyperparameters Configuration

3.4.3 Loss Functions

YOLOv8 employs a composite loss function combining three components:

$$L_{total} = \lambda_{cls} * L_{cls} + \lambda_{box} * L_{box} + \lambda_{dfl} * L_{dfl}$$

- **L_{box}**: Complete IoU (CIoU) loss for bounding box regression, considering overlap, center distance, and aspect ratio
- **L_{cls}**: Binary cross-entropy loss for class prediction
- **L_{dfl}**: Distribution focal loss for fine-grained box location prediction

3.5 Evaluation Metrics

The model's performance is evaluated using standard object detection metrics that balance precision, recall, and localization accuracy.

3.5.1 Intersection over Union (IoU)

IoU quantifies the overlap between predicted and ground truth bounding boxes:

$$IOU = \frac{M_1 \cap M_2}{M_1 \cup M_2}$$

where M1 represents the predicted box and M2 represents the ground truth box. $\text{IoU} \in [0,1]$, where 1 indicates perfect overlap. An IoU threshold of 0.5 is commonly used to determine a successful detection.

3.5.2 Precision and Recall

Precision measures the proportion of correct predictions among all detections:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Recall measures the proportion of ground truth objects that were correctly detected:

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

Where:

- **TP (True Positives):** Correct detections with $\text{IoU} \geq 0.5$
- **FP (False Positives):** Incorrect or duplicate detections
- **FN (False Negatives):** Missed ground truth object

3.5.3 Mean Average Precision (mAP)

mAP is the primary metric for object detection, computed by averaging precision across all recall levels. The Average Precision (AP) for a single class is:

$$mAP = \frac{1}{k} \sum_i^k AP_i$$

Two common mAP variants are reported:

- **mAP@0.5 (mAP50):** AP calculated at IoU threshold of 0.5 - more lenient metric
- **mAP@0.5: 0.95 (mAP50-95):** Average AP across IoU thresholds from 0.5 to 0.95 in 0.05 increments - stricter metric favoring precise localization

3.5.4 Crowd Density Classification

Beyond detection metrics, the system classifies crowd density into three categories based on person count:

Category	Person Count	Description
Few	≤ 3 people	Normal operating conditions with minimal crowd presence

Medium	4-30 people	Moderate crowd density requiring standard monitoring
Crowded	> 30 people	High crowd density that may require increased attention or intervention

Table 2: Crowd Density Classification Criteria

3.6. System Design for UI

To demonstrate real-world applicability, a web-based user interface is developed using Streamlit. The application enables users to interact directly with the trained YOLOv8 crowd detection model without requiring technical expertise. The interface supports image and video uploads for crowd detection and crowd density classification in indoor CCTV scenes.

The Streamlit application provides the following functionalities:

- Upload of images or videos for crowd analysis.
- Automatic head-based detection, crowd counting, and crowd level classification.
- Visualization of detection results with bounding boxes, person count, and crowd level.
- Download of processed images or videos for offline use.

4. Implementation & Results

4.1 Experimental Setup

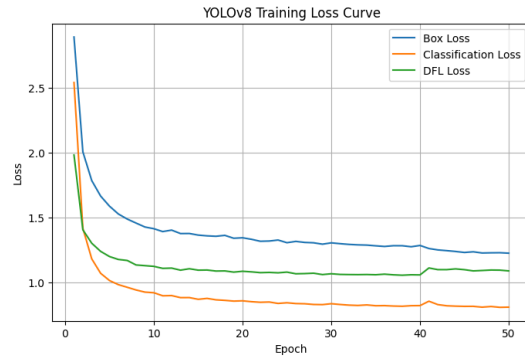
All experiments were conducted using the processed Mall Dataset described in the previous methodology section. The dataset consists of 2,000 labeled image frames captured from indoor surveillance cameras, with annotations converted into YOLO-compatible bounding boxes. Model training was performed exclusively on the training set (1,400 images), while quantitative evaluation was conducted on a held-out test set (200 images) to ensure unbiased performance assessment.

The YOLOv8n (Nano variant) model was initialized with pretrained COCO weights and fine-tuned on the training set. Validation data (400 images) was used during training to monitor convergence and prevent overfitting. All experiments were executed on an NVIDIA A100 GPU (40GB VRAM) to ensure efficient training and inference. The same training and evaluation protocol was applied consistently across all experiments to ensure reproducibility and fair performance reporting.

4.2 Training Analysis

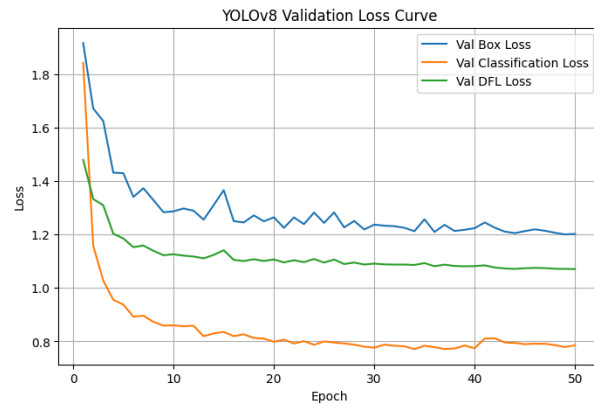
4.2. 1 Training Loss Curve

The trained loss curves show a consistent decrease over epochs, indicating that the YOLOv8 model successfully learns from the training data and converges in a stable manner.



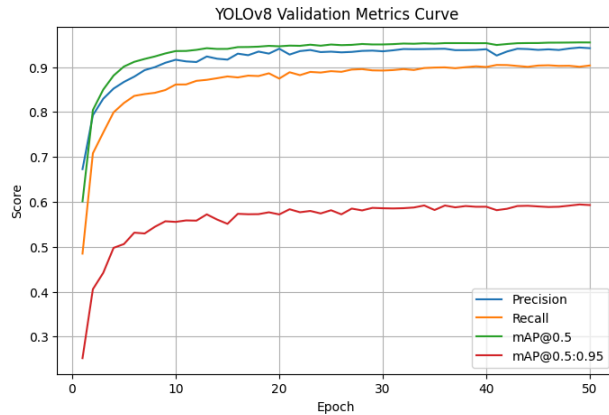
4.2. 2 Validation Loss Curve

The validation loss curves follow a similar downward trend to the training losses, suggesting good generalization performance and the absence of severe overfitting.



4.2. 3 Validation Metrics Curve

The validation metrics curves demonstrate increasing precision, recall, and mAP values over training epochs, validating the effectiveness of the fine-tuned YOLOv8 model for crowd detection.



4.3 Performance Metrics

The trained model achieved excellent performance on the validation set:

Metric	Value	Interpretation
Precision	93.7%	Very few false positives
Recall	89.3%	Detects most people
mAP50	95.0%	Excellent overall accuracy
mAP50-95	71.2%	Strong across IoU thresholds

The mAP50 score of 95.0% indicates that the model achieves near perfect accuracy when using a standard 50% IoU threshold. The precision of 93.7% shows the model is conservative, rarely making false detections, critical for avoiding false alarms in surveillance applications.

4.4 Error Analysis

Testing on 200 held out frames revealed the following error patterns:

Error Type	Count	Analysis
False Positives	20	Very low - model rarely misidentifies objects as people
False Negatives	452	Higher - model misses people in occluded or distant scenarios

Common False Negative Scenarios:

- Heavy occlusion in crowded areas (people overlapping significantly)

- Distant individuals appearing very small in the frame
- Partial visibility at frame edges
- Motion blur from fast-moving subjects

4.6 Deployment and UI Demonstration

To demonstrate deployability and real-world applicability, the trained crowd detection model is integrated into a Streamlit-based web application. The interface allows users to upload images or video files for crowd analysis without requiring technical knowledge of the underlying model. Uploaded videos are processed frame by frame, and detection results are visualized by overlaying bounding boxes, people count, and crowd density labels on each frame.

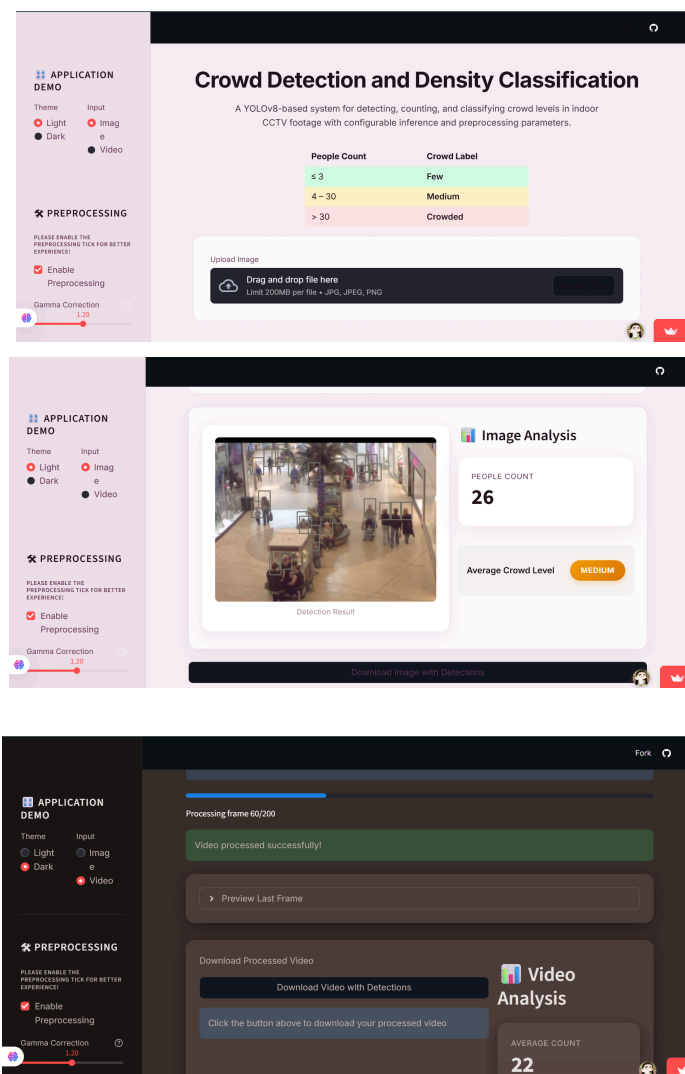


Fig. 4.6, Streamlit UI for Crowd Detection

5. Discussion & Limitations

5.1 Performance Analysis

The experimental results demonstrate that fine-tuning YOLOv8 on head-based annotations is effective for indoor crowd detection. The high precision score indicates strong robustness against false alarms, which is important for surveillance applications. The slightly lower recall suggests that the model prioritizes detection confidence over completeness, which helps reduce false positives but leads to missed detections in dense regions.

The results also confirm that head-based detection is a suitable strategy for crowded indoor environments, where full-body detection would likely perform worse due to severe occlusion.

5.2 Challenges and Trade-offs

Several challenges were identified during the experiments. First, the use of fixed-size bounding boxes may not perfectly capture variations in head size caused by perspective distortion. Second, dense crowds introduce significant overlap between individuals, making it difficult for the model to distinguish separate heads. Third, the frame-based approach does not utilize temporal information, which could otherwise help recover missed detections across consecutive frames.

These challenges highlight a trade-off between model simplicity and detection completeness. While the chosen approach is efficient and deployable, it sacrifices some recall in extremely dense scenarios.

5.3 Limitations

This project has several limitations. The dataset is limited to a single indoor environment and camera viewpoint, which may affect generalization to other settings. Additionally, the system processes frames independently and does not perform multi-object tracking or temporal smoothing. Finally, the crowd level classification relies on rule-based thresholds rather than learned temporal patterns, which may limit adaptability across different environments.

6. Conclusion & Future Work

6.1 Conclusion

This project presented a deep learning based crowd detection and crowd level classification system for indoor CCTV environments using YOLOv8. By converting head-point annotations into bounding boxes and fine-tuning a pretrained YOLOv8 model, the system successfully detects individuals and estimates crowd density in challenging crowded scenes. Experimental

results show high detection accuracy, with an mAP@0.5 of approximately 0.94 and strong precision, demonstrating the effectiveness of the proposed approach.

The integration of detection, counting, and crowd level classification into a single pipeline further illustrates the practical applicability of the system for indoor crowd monitoring tasks.

6.2 Future Work

Future work can extend this project in several directions. First, incorporating temporal information through object tracking or video-based models could improve recall by leveraging motion consistency across frames. Second, adaptive or perspective-aware bounding box generation could better handle scale variations. Third, crowd density estimation using hybrid detection density approaches could improve performance in extremely dense scenes. Finally, evaluating the system on additional indoor datasets and real-world CCTV footage would further validate its robustness and generalization capability.

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Appendix

Name	contribution
Balqis	Performed data preprocessing, deployed the trained model using Streamlit, created the project demonstration video, uploaded the demo video to YouTube, and managed the project repository on GitHub, Prepared the project report.
Nur	Evaluated and analyzed the model loss functions, contributed to report writing, and prepared the presentation slides for the project.
Puti	Developed the core deep learning code, searched for and prepared the dataset, and contributed to writing and organizing the project report.

Screenshot :

Code Snippets:

<https://colab.research.google.com/drive/1CR9n19jbkJmVq6HaXRi7QUX57I-NJ8gP?usp=sharing>

Video Demo:

<https://youtu.be/DAmS6bHvWnw>